

Requirement Analysis

1. Customer Journey Map

This map follows a **System Administrator or Data Analyst** persona whose goal is to keep target tables (like the CMDB or User table) up-to-date without manual entry.

Stage	User Action	Thinking / Feeling	Touchpoint
Preparation	Clean external CSV/Excel file.	"Is the data header format correct?"	Excel / CSV
Configuration	Create a Data Source and Load Data.	"Hope the staging table builds correctly."	ServiceNow Data Source
Mapping	Create Transform Map with Dot Walking.	"Need to map 'Dept' to 'User.Department.Name'."	Transform Map Assist
Automation	Set up a Scheduled Import.	"This should run every Monday at 2 AM."	Scheduled Import

2. Data Flow Diagram (DFD): The Import Lifecycle

To visualize how data moves through the system, here is the structured flow of information:

Code snippet

graph TD

A[External Source] -->|Raw Data: CSV, API, etc.| B[Staging Area]

B -->|Import Set Table| C[Transformation Engine]

C -->|Transform Map & Dot Walking| D[Target Table]

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- **Step 1: External Source:** Raw data (CSV, JDBC, API) is sent to ServiceNow.
- **Step 2: Staging Area:** Data is temporarily stored in an Import Set Table.
- **Step 3: Transformation Engine:** The Transform Map applies logic, using Dot Walking to resolve reference relationships (e.g., matching a Manager's Name to their sys_id).
- **Step 4: Target Table:** Clean, mapped data is inserted or updated in production tables (such as sys_user or cmdb_ci).

3. User Stories

- **As a System Admin,** I want to use dot walking in transform scripts so that I can map incoming strings to reference fields in related tables automatically.
- **As a Data Owner,** I want to schedule imports so that my records remain synchronized with the source system without manual intervention.

4. Solution Requirements

Functional Requirements

- **Coalesce Execution:** The system must support Coalesce on unique identifiers (like Employee ID) to prevent duplicate records.
- **Scripted Field Maps:** The system must allow Scripted Field Maps for complex relationship resolution where simple auto-mapping fails.

Non-Functional Requirements

- **Scalability:** The system must process imports of \$10,000\$+ records within a \$15\$-minute window.
- **Security:** Ensure the Service Account used for imports has minimum viable permissions (e.g., import_admin role).

5. Technology Stack (Architecture)

- **Platform:** ServiceNow.
- **Data Sources:** JDBC, SFTP, or REST API.
- **Logic Layer:** JavaScript (GlideRecord and GlideAjax) for transform scripts.
- **Storage:** ServiceNow DB (MariaDB/MySQL backend) using Staging (Import Set) Tables.